MATH 332 Linear Algebra: Theory and Applications

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Instructions:

i. *This test has **THREE questions**. Attempt all. The maximum number of points is **100**.*

ii. *The time allowed is 75 minutes.*

iii. *This test is closed book, but you may use **4** double-sided letter-size sheets of notes.*

iv. *No calculators or other electronic devices are allowed. Phones must be placed in your bags under your desks or at the front of the room. Hands must be on top of your desks.*

v. *You are expected to simplify any expressions that can be simplified as whole numbers, e.g.,* $36/15 = 12/5$*, but you may leave other expressions as is, e.g.,* $\sqrt{47}$ *or* $18/29$*.*

vi. *Show all your work to justify your answers. Answers without adequate justification will not receive credit.*

vii. *Print your name clearly on each answer sheet.*

viii. *Off-site students may contact the instructor as directed by your syllabus.*

Q	Max	Score
1	30	
2	30	
3	40	
Tot	**100**	

ChatGPT/Codex assisted in the construction of this exam.

I understand these instructions and have not relied on any help for this exam beyond what is allowed, nor provided any help to anyone else beyond what is allowed.

_____ _____

Signature *Date*

_____

Printed Name

1. (30 points) Let S be the unit square (including its interior) with vertices $(0,0)$, $(1,0)$, $(1,1)$, and $(0,1)$. For each matrix, A_i, below, let $T_i = \{\mathsf{A}_i\boldsymbol{x} : \boldsymbol{x} \in S\}$ be the transformed set of points

   i. Identify what kind of shape T_i is.

   ii. Find the area of T_i.

   iii. Compute $\det(\mathsf{A}_i)$.

   iv. Explain the relationship between the area of T_i and $\det(\mathsf{A}_i)$.

   a. (10 points) $\mathsf{A}_1 = \begin{bmatrix} -2 & 0 \\ 0 & 3 \end{bmatrix}$.

   Answer:

   i. T_1 *is the rectangle with vertices* $(0,0)$, $(-2,0)$, $(-2,3)$, *and* $(0,3)$, *since* $\mathsf{A}_1(x,y)^T = (-2x, 3y)^T$.

   ii. *Its side lengths are* 2 *and* 3, *so* $\text{area}(T_1) = 2 \cdot 3 = \boxed{6}$.

   iii. $\det(\mathsf{A}_1) = (-2)(3) - 0 \cdot 0 = \boxed{-6}$.

   iv. $\text{area}(T_1) = |\det(\mathsf{A}_1)| = 6$. *The negative determinant indicates reversed orientation: the map expands in both directions and reflects across the* y*-axis.*

   b. (10 points) $\mathsf{A}_2 = \begin{bmatrix} 2 & 1 \\ 4 & 5 \end{bmatrix}$.

   Answer:

   i. T_2 *is the nonrectangular parallelogram with vertices* $(0,0)$, $(2,4)$, $(3,9)$, *and* $(1,5)$. *Its adjacent edge vectors are* $(2,4)^T$ *and* $(1,5)^T$*; their dot product is* $22 \neq 0$, *so they are not perpendicular.*

   ii. *The parallelogram area is* $\text{area}(T_2) = |2 \cdot 5 - 1 \cdot 4| = \boxed{6}$.

   iii. *By* $ad - bc$, $\det(\mathsf{A}_2) = 2 \cdot 5 - 1 \cdot 4 = \boxed{6}$. *Alternatively,* $R_2 \leftarrow R_2 - 2R_1$ *gives* $\left[\begin{smallmatrix} 2 & 1 \\ 0 & 3 \end{smallmatrix}\right]$ *without changing the determinant, so the diagonal product is* $2 \cdot 3 = 6$.

   iv. $\text{area}(T_2) = |\det(\mathsf{A}_2)| = 6$. *Here the determinant is positive, so orientation is preserved.*

   c. (10 points) $\mathsf{A}_3 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$.

   Answer:

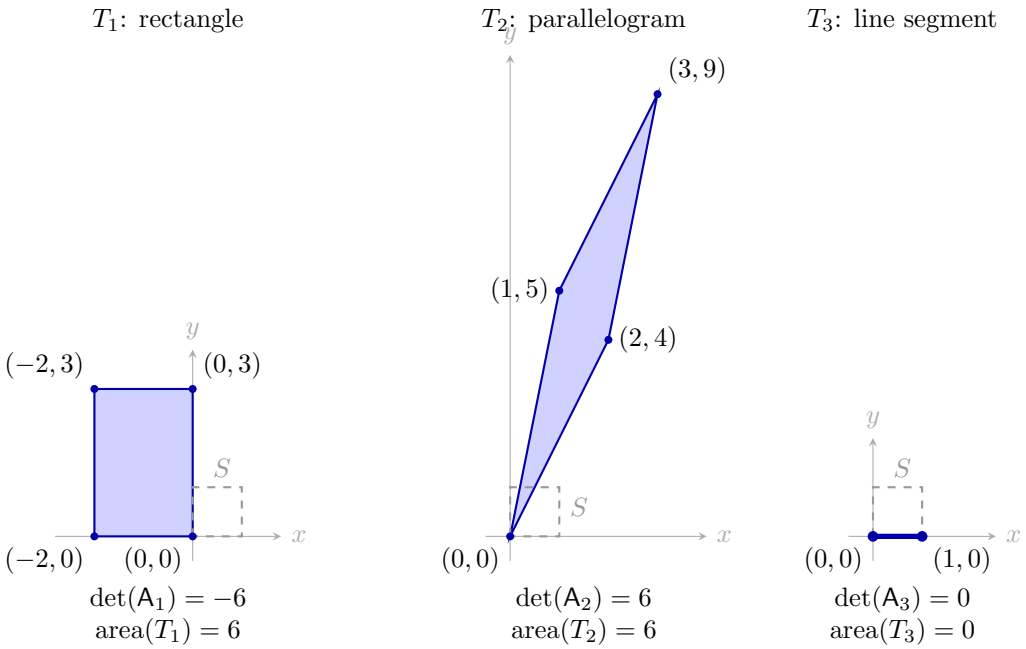
   i. T_3 *is the line segment from* $(0,0)$ *to* $(1,0)$, *since* $\mathsf{A}_3(x,y)^T = (x,0)^T$ *projects onto the* x*-axis.*

   ii. $\text{area}(T_3) = \boxed{0}$. *The segment has length* 1, *but zero two-dimensional area.*

   iii. $\det(\mathsf{A}_3) = 1 \cdot 0 - 0 \cdot 0 = \boxed{0}$.

   iv. $\text{area}(T_3) = |\det(\mathsf{A}_3)| = 0$*: the square collapses to a line segment.*

Problem 1: images of the unit square



Dashed outlines show S. All three panels use the same unit scale.

2. (30 points) In $\mathbb{R}^3$, let

$$\boldsymbol{p} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \qquad \boldsymbol{q} = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}, \qquad \boldsymbol{u} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix},$$

$$L = \{\boldsymbol{p} + t\boldsymbol{u} : t \in \mathbb{R}\}, \qquad V = \operatorname{span}\{\boldsymbol{u}\}.$$

Use the usual dot product and Euclidean length.

a. (20 points) Find the distance from $\boldsymbol{q}$ to L.

Answer: With $\boldsymbol{w} = \boldsymbol{q} - \boldsymbol{p} = (1, 1, 0)^T$, we have $\boldsymbol{u}^T\boldsymbol{w} = 3$ and $\boldsymbol{u}^T\boldsymbol{u} = 9$. The closest point on L is therefore

$$\boldsymbol{r} = \boldsymbol{p} + \frac{\boldsymbol{u}^T\boldsymbol{w}}{\boldsymbol{u}^T\boldsymbol{u}}\boldsymbol{u} = \boldsymbol{p} + \frac{1}{3}\boldsymbol{u} = \begin{bmatrix} 4/3 \\ 2/3 \\ 5/3 \end{bmatrix}.$$

The residual and its length are

$$\boldsymbol{q} - \boldsymbol{r} = \begin{bmatrix} 2/3 \\ 1/3 \\ -2/3 \end{bmatrix}, \qquad \operatorname{dist}(\boldsymbol{q}, L) = \|\boldsymbol{q} - \boldsymbol{r}\| = \sqrt{\frac{4 + 1 + 4}{9}} = \boxed{1}.$$

Indeed, $\boldsymbol{u}^T(\boldsymbol{q} - \boldsymbol{r}) = (2 + 2 - 4)/3 = 0$, so the residual is perpendicular to the line.

Alternative: minimize the distance over the line parameter. *A general point on L is $\boldsymbol{p} + t\boldsymbol{u} = (1 + t, 2t, 1 + 2t)^T$, so its distance from $\boldsymbol{q}$ is*

$$d(t) = \|\boldsymbol{q} - (\boldsymbol{p} + t\boldsymbol{u})\| = \sqrt{(1 - t)^2 + (1 - 2t)^2 + (-2t)^2} = \sqrt{9t^2 - 6t + 2}.$$

Minimizing $d(t)$ is equivalent to minimizing its square. Since

$$d(t)^2 = (3t - 1)^2 + 1 \geq 1,$$

the global minimum occurs at $t = 1/3$, giving $\boxed{\operatorname{dist}(\boldsymbol{q}, L) = 1}$. *Completing the square is a useful trick: it writes the expression as a nonnegative square plus a constant, making both the minimum value and where it occurs immediately visible.*

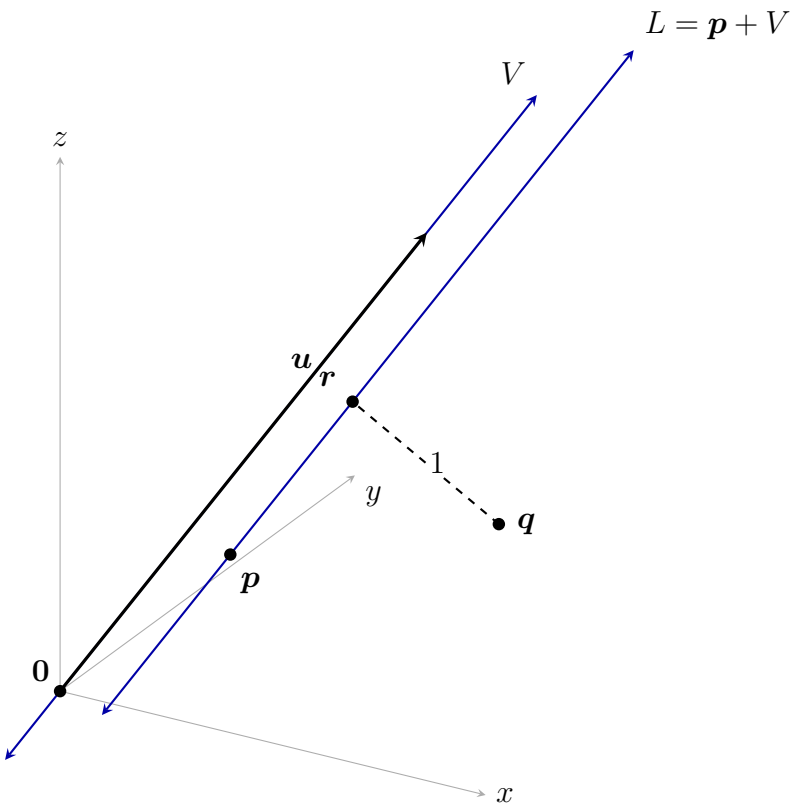
b. (10 points) Are L and V subspaces of $\mathbb{R}^3$ under the usual operations? What is the relationship between L and V?

Answer: L is not a subspace because it does not contain $\mathbf{0}$: $\boldsymbol{p} + t\boldsymbol{u} = \mathbf{0}$ requires $t = 0$ from the second component, but then the first component is 1. The set V is a subspace: it contains zero, and every linear combination of multiples of $\boldsymbol{u}$ is again a multiple of $\boldsymbol{u}$. Moreover,

$$\boxed{L = \boldsymbol{p} + V = \{\boldsymbol{p} + \boldsymbol{v} : \boldsymbol{v} \in V\}}.$$

Thus L is a translate of V. They are distinct parallel lines: V passes through the origin, and L passes through $\boldsymbol{p}$.

Problem 2: geometric sketch



3. (40 points) Work in the real vector space $\mathbb{P}_2$ of polynomials of degree at most two, with the usual addition and scalar multiplication. Define

$$p_1(t) = 1 - t, \qquad p_2(t) = t - t^2, \qquad p_3(t) = 1 - \lambda t^2, \qquad \lambda \in \mathbb{R}.$$

Give reasons for all answers.

a. (15 points) For which values of λ do p_1, p_2, p_3 form a basis of $\mathbb{P}_2$?

Answer: Matching coefficients in $c_1p_1 + c_2p_2 + c_3p_3 = 0$ *gives*

$$c_1 + c_3 = 0, \qquad -c_1 + c_2 = 0, \qquad -c_2 - \lambda c_3 = 0.$$

Hence $c_1 = c_2 = -c_3$ *and* $(1-\lambda)c_3 = 0$. *If* $\lambda \neq 1$, *all coefficients are zero, so the polynomials are independent. Since* $\dim \mathbb{P}_2 = 3$, *three independent polynomials form a basis. If* $\lambda = 1$, $p_3 = p_1 + p_2$, *so the set is dependent and is not a basis. Thus the family is a basis exactly when* $\boxed{\lambda \neq 1}$.

b. (10 points) Now set $\lambda = 1$ and let $V = \operatorname{span}\{p_1, p_2, p_3\}$. Find a basis of V and its dimension.

Answer: Since $p_3 = p_1 + p_2$, $V = \operatorname{span}\{p_1, p_2\}$. *The equation* $ap_1 + bp_2 = 0$ *has constant coefficient* $a = 0$ *and* t^2 *coefficient* $-b = 0$. *Thus* $\boxed{\{1 - t,\ t - t^2\} \text{ is a basis of } V}$ *and* $\boxed{\dim V = 2}$.

Comment: *Some responses gave the coordinate vectors of the polynomials relative to the basis* $\{1, t, t^2\}$ *rather than the polynomials themselves. The coordinate map gives a one-to-one linear correspondence, but a polynomial and its coordinate vector are different objects. A basis of* V *must consist of polynomials in* V; *their coordinate vectors form a basis of the corresponding subspace of* $\mathbb{R}^3$.

c. (5 points) Is $V = \mathbb{P}_2$? If not, find a polynomial $q \in \mathbb{P}_2$ that is not in V. Justify your answer.

Answer: No: $\dim V = 2$, *whereas* $\dim \mathbb{P}_2 = 3$. *For example, take* $\boxed{q(t) = 1}$. *Both* $p_1(1) = 0$ *and* $p_2(1) = 0$, *so every polynomial in* $V = \operatorname{span}\{p_1, p_2\}$ *vanishes at* 1. *Since* $q(1) = 1 \neq 0$, $q \notin V$.

d. (10 points) Let $W = \{p \in \mathbb{P}_2 : p(0) = p(1)\}$. Is W a subspace of $\mathbb{P}_2$?

Answer: Yes. The zero polynomial belongs to W. *For* $f, g \in W$ *and* $a, b \in \mathbb{R}$, *let* $h = af + bg$. *Then* $h \in \mathbb{P}_2$ *and*

$$h(0) = af(0) + bg(0) = af(1) + bg(1) = h(1).$$

Thus $h \in W$, *so* W *is closed under linear combinations and is a subspace of* $\mathbb{P}_2$.

Test 2 Scores (out of 100)

Number of Students: 18, Minimum: 20, Maximum: 99, Mean: 63.7, Median: 63
Standard Deviation: 23.5, Quartiles (Q1, Q3): (48, 80)

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9|1379
8|0
7|55
6|0245
5|36
4|88
3|2
2|08
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